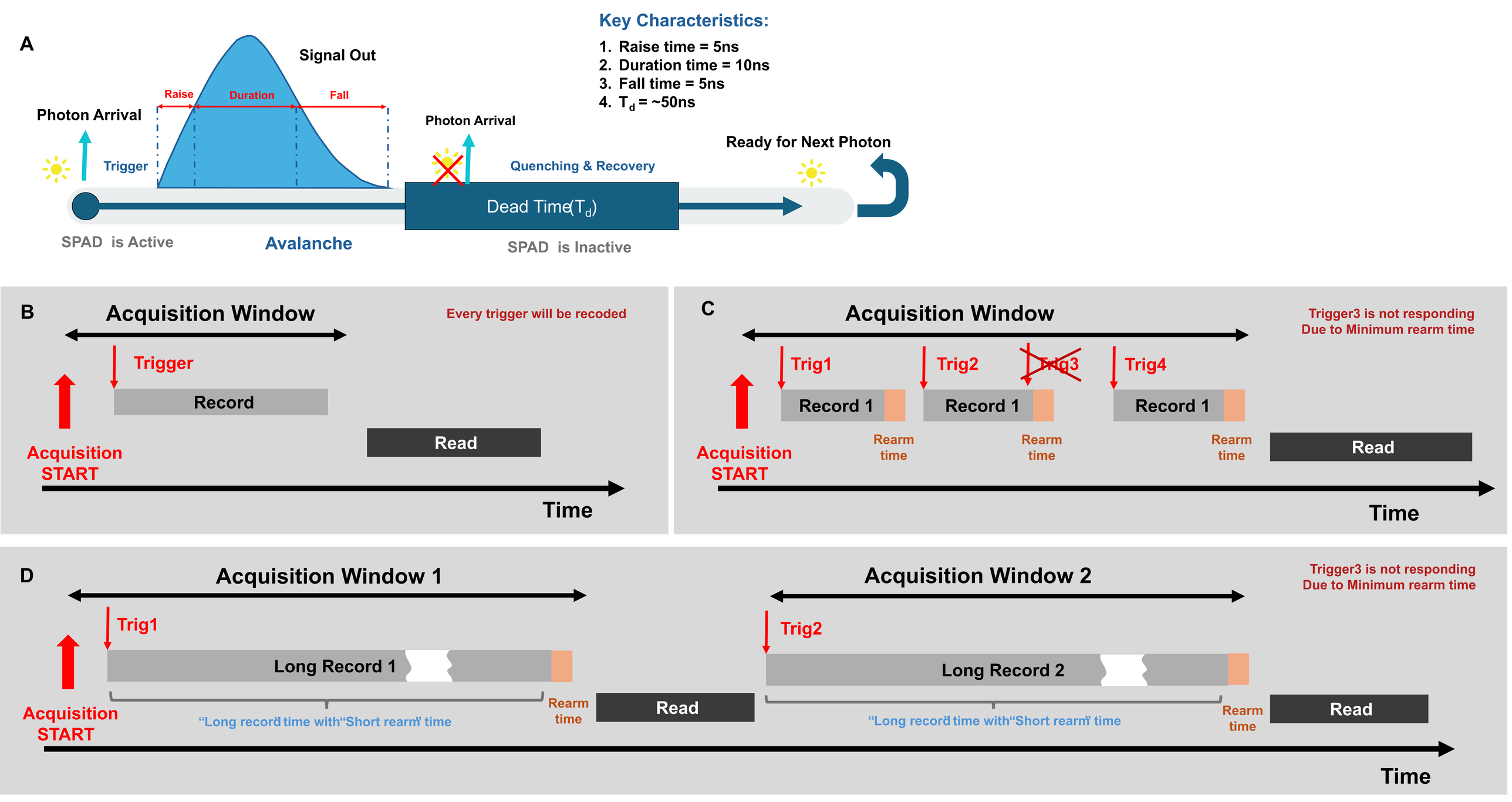
A
Signal Out
Raise
Duration
Fall
Photon Arrival
Trigger
Photon Arrival
Quenching & Recovery
Ready for Next Photon
Dead Time(T_d)
SPAD is Active
Avalanche
SPAD is Inactive
Key Characteristics:
1. Raise time = 5ns
2. Duration time = 10ns
3. Fall time = 5ns
4. T_d = ~50ns
B
Acquisition Window
Every trigger will be recoded
Trigger
Record
Read
Acquisition START
Time
C
Acquisition Window
Trigger3 is not responding Due to Minimum rearm time
Trig1
Trig2
Trig3
Trig4
Record 1
Record 1
Record 1
Rearm time
Rearm time
Rearm time
Read
Acquisition START
Time
D
Acquisition Window 1
Trig1
Long Record 1
"Long record" time with "Short rearm" time
Rearm time
Read
Acquisition START
Acquisition Window 2
Trigger3 is not responding Due to Minimum rearm time
Trig2
Long Record 2
"Long record" time with "Short rearm" time
Rearm time
Read
Time